Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Hypothesis Testing

SOLUTIONS

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1. $H_0 : p = 0.5$; $H_1 : p > 0.5$.
2. Probability of rejecting H_0 when it is true (Type I error rate).
3. Rejecting H_0 when it is true.
4. Failing to reject H_0 when it is false.
5. Find smallest c with $P(X \geq c | p = 0.5) \leq 0.05$; CR $X \geq 9$ (check tables).
6. 5% chance of rejecting true H_0.
7. Reject for unusually small or large X; split α both tails.
8. $P(X \geq 8 | p = 0.5)$; compare with 0.05; conclude.
9. $np > 5$, $n(1-p) > 5$ (or similar rule).
10. One-tailed: direction specified; two-tailed: either direction.
11. $p = 0.03 < 0.05 \Rightarrow$ reject H_0; significant evidence mean < 50.
12. $H_0 : \mu = 100$; $H_1 : \mu > 100$.
13. $P(X \leq 2 | n = 15, p = 0.3)$.
14. Absence of evidence is not evidence of absence; data may be insufficient.
15. $Z = \frac{48-50}{4/\sqrt{20}}$; compare with -1.645; conclude.
16. $P(X \leq 10 | n = 25, p = 0.6)$; if ≤ 0.05 reject.
17. Smallest c with $P(X \geq c) \leq 0.05$.
18. $Z = \frac{498-500}{5/\sqrt{25}} = -2$; reject at 5% (critical -1.645).
19. Probability, under H_0, of result at least as extreme as observed.
20. Normal approx $X \sim N(20, 10)$; continuity; two-tailed test.
21. $H_0 : p = 0.5$, $H_1 : p \neq 0.5$; sig level $= P(X \leq 3) + P(X \geq 12)$.
22. $H_0 : p = 0.9$, $H_1 : p < 0.9$; $P(X \leq 40 | n = 50, p = 0.9)$.
23. Critical region: reject if statistic in region; p-value: reject if $p \leq \alpha$.
24. $Z = \frac{22-20}{3/\sqrt{36}} = 4$; reject at 1% ($z_{0.99} \approx 2.33$).
25. Avoid data-driven choice of α that inflates Type I error.